UDP Header

	0	1	2	3
0	Source Port		Destination Port	
4	Length		Checksum	

Common UDP Ports

7	echo	137	netbios-ns	546	DHCPv6c
19	chargen	138	netbios	547	DHCPv6s
53	domain	161	snmp	1900	SSDP
67	DHCPc	162	snmp-trap	5353	mDNS
68	DHCPs	500	isakmp		
69	tftp	514	syslog		
123	ntp	520	Rip		

Length: number of bytes including UDP header.
Minimum value is 8
Checksum includes pseudo-header (IPs, length, protocol), UDP header and payload.

ARP

	0	1	2	3
0	HW Addr. Type		Prot. Addr. Type	
4	HW Addr Len.	Prot. Addr Len	Opcode	
8	Source Hardware Addr.			
12	Src HW Addr		Src Protocol Addr	
16	Src. Proto Addr		Tgt HW Addr	
20	Tgt HW Address (cont.)			
24	Target Protocol Address			

Hardware Type: 1 - Ethernet
Protocol Type: 0x0800 - IPv4
Address Length: 4=IPv4, 6=Ethernet
Opcode: 1-request, 2-response

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Ethernet Frame: 14 Bytes
Dest Addr (6), Src Addr (6) Type (2)

SANS

TCP/IP and tcpdump

Version June 2016

POCKET REFERENCE GUIDE

Please submit comments and corrections to jullrich@sans.edu
https://www.sans.org/security-resources/tcpip.pdf

COURSES & GIAC CERTIFICATIONS

SEC503
Intrusion Detection In-Depth — GCIA

SEC 401
Security Essentials — GSEC

SEC 502
Perimeter Protection — GPPA

SEC 560
Network Penetration Testing — GPEN

SEC 546
IPv6 Security Essentials — IPv6 Forum Certified Course

FOR 572
Network Forensics — GNFA

MGT512
Security Leadership Essentials — GSLC

tcpdump usage

tcpdump [-aAenStvxX] [-F filterfile] [-i int] [-c n]
[-r pcapfile] [-s snaplen] [-w pcapfile] ['bpf filter']

-A	display payload
-c n	display first n packets
-D	list interfaces
-e	display data link header
-F	read filter expression from file
-i	listen on specified interface
-n	do not resolve IP addresses / ports
-r	read packets from file
-s	set snap length in bytes
-S	display absolute TCP sequence numbers
-t	do not print timestamp
-tttt	print date and time
-v	verbose (multiple v: more verbose)
- w	write packets to file
-x	display in hex
-xx	display link layer in hex
-X	display in hex + ASCII

Acronyms

AH	Authentication Header (RFC 2402)
ARP	Address Resolution Protocol (RFC 826)
BGP	Border Gateway Protocol (RFC 1771)
CWR	Congestion Window Reduced (RFC 2481)
DF	Do not fragment flag (RFC 791)
DHCP	Dynamic Host Configuration Protocol (RFC 2131)
DNS	Domain Name System (RFC 1035)
ECN	Explicit Congestion Notification (RFC 3168)
ESP	Encapsulating Security Payload (RFC 2406)
FTP	File Transfer Protocol (RFC 959)
GRE	Generic Route Encapsulation (RFC 2784)
HTTP	Hypertext Transfer Protocol (RFC 1945)
ICMP	Internet Control Message Protocol (RFC 792)
IGMP	Internet Group Management Protocol (RFC 2236)
IMAP	Internet Message Access Protocol (RFC 2060)
IP	Internet Protocol (RFC 791)
ISAKMP	Internet Sec. Assoc. & Key Mngm Proto. (RFC 7296)
L2TP	Layer 2 Tunneling Protocol (RFC 2661)
OSPF	Open Shortest Path First (RFC 1583)
POP3	Post Office Protocol v3 (RFC 1460)
RFC	Request for Comments
SMTP	Simple Mail Transfer Protocol (RFC 821)
SSH	Secure Shell (RFC 4253)
SSL	Secure Sockets Layer (RFC 6101)
TCP	Transmission Control Protocol (RFC793)
TLS	Transport Layer Security (RFC 5246)
TFTP	Trivial File Transfer Protocol (RFC 1350)
TOS	Type of Service (RFC 2474)
UDP	User Datagram Protocol (RFC 768)

DNS

	0	1	2	3
0	Query ID		Flags (see below)	
4	Query Count		Answer Count	
8	Authority Rec. #		Addtl. Record #	
12	Questions… Answers… Authority Records… Additional Records…			

Flags:

Byte Offset 2								Byte Offset 3							
0	1	2	3	4	5	6	7	0	1	2	3	4	5	6	7
QR	OPCODE				AA	TC	RD	RA	Z	AD	CD	RCODE			

QR: Query (0) or Response (1)
Opcode: 0 – std. Query, 1 – inverse query (IQUERY), 2 – Server Status (STATUS)
AA: Authoritative Answer
TC: Truncated response
RD Recursion Desired
RA Recursion Available
Z Zero (set to 0)
AD Authentic Data(DNSSEC)
CD Checking Disabled (DNSSEC)

RCODE:
0 – No error
1 – Format Error
2 – Server Failure
3 – Non-existent domain (NXDOMAIN)
4 – Query type not implemented
5 – Query refused

ICMP

	0	1	2	3
0	Type	Code	Checksum	
4	Addtl. information depending on type/code			

Type	Code	Name
0	0	Echo Reply
3	0	Network Unreachable
	1	Host Unreachable
	2	Protocol Unreachable
	3	Port Unreachable
	4	Fragmentation Required
	5	Source Route Failed
	6	Dest. Network Unknown
	7	Destination Host Unknown
	8	Source Host Isolated
	9	Net Administratively Prohibited
	10	Host Administratively Prohibited
	11	Network unreachable for TOS
	12	Host unreachable for TOS
	13	Communication Admin. Prohibited
4	0	Source quench
5	0	Network Redirect
	1	Host Redirect
	2	ToS & Network Redirect
	3	ToS & Host Redirect
8	0	Echo [Echo Request]
9	0	Router Advertisement
11	0	Time to live exceeded in transit
	1	Fragment Reassembly time exceeded
12	0	Parameter Prob. Pointer indicated the error
	1	Missing a required option
	2	Bad length
13	0	Timestamp
14	0	Timestamp Reply
15	0	Information Request
16	0	Information Reply
17	0	Address Mask Request
18	0	Address Mask Reply
30	0	Traceroute

ICMP Echo Request/Reply (Ping)

	0	1	2	3
0	Type	Code	Checksum	
4	ICMP ID		ICMP Sequence	

IPv4 Header

Offset: Add column+row. e.g. Protocol=9
ip[9] = "IP header offset 9" or the protocol field

	0		1	2		3
0	Ver	IHL	TOS	Total Length		
4	IP Identification			X D M	Offset	
8	TTL		Protocol	Checksum		
12	Source Address					
16	Destination Address					
20	Options (optional)					

Version: 4 ip[0]&0xf0
Header Length: IP header length in double-words (4 bytes). Minimum 5 (20 bytes)
ToS/Differentiated Services Byte ip[1]

0	1	2	3	4	5	6	7
Diff. Svc. Code Point						ECN	

Total Length: includes header ip[2:2]
Flags ip[6]

0	1	2	3	4	5	6	7
X	D	M	O	O	O	O	O

X: Reserved, D: Do Not Frag. M: More Fragments
O: Offset bits
Fragment Offset: position of this ip datagram's payload in original packet (multiply by 8)
Protocol ip[9]

1	ICMP	17	UDP	50	ESP
2	IGMP	41	IPv6	51	AH
6	TCP	47	GRE	115	L2TP

Checksum: IP Header Only

Options: up to 40 bytes, 4 byte padded ip[20..]

0	End of Options List	68	Timestamp
1	No Operation	131	Loose source route
7	Record Route	137	Strict Source Route

TCP

	0	1	2	3
0	Source Port		Dest. Port	
4	Sequence Number			
8	Acknowledgement Number			
12	HL	R	Flags	Window Size
16	Checksum		Urgent Pointer	
20	Options (up to 40 bytes)			

Common TCP Ports

20	ftp-data	80	http	443	https
21	ftp	88	kerberos	445	MS SMB
22	ssh	110	pop3	465	SMTPS
23	telnet	113	authd	1433	MS SQL
25	smtp	119	nntp	3128	Squid
43	whois	143	imap	3306	Mysql
53	dns	179	bgp	3389	MS Term.

Advanced

- Sequence Number tcp[4:4]: increments with each byte
- Ack. Number tcp[8:4]: next expected sequence number
- Header Length tcp[12]>>4: TCP Header Length / Offset; minimum 5. Number of 32 bit dwords (4 bytes)
- Reserved tcp[12]&0x0f: Set to 0
- Flags tcp[13]

7	6	5	4	3	2	1	0
8	4	2	1	8	4	2	1
CWR	ECE	URG	ACK	PUSH	RES	SYN	FIN

Window Size tcp[14:2]: recv. Window size
Checksum tcp[16:2]: Covers pseudo-header + TCP Header + TCP Payload
Urgent Point tcp[18:2]: Offset pointer to urgent data
Options tcp[20:..]

0	End of List	3	Window Scale
1	No Operation	4	Selective Ack OK
2	Max. Segment Size	8	Timestamp
29	TCP Auth Option	30	Multipath TCP